

Task 1: Python File Handling And Automation

OBJECTIVE:

The objective of this task is to learn Python file handling, automate file operations like reading, writing, renaming, moving, and deleting files, and implement error handling using try-except blocks.

Explanation

CODE SNIPPETS:

Filename: main.py

Import required modules

import os

import shutil

Function to read a text file

def read_file(filename):

try:

with open(filename, "r") as file:

content = file.read()

print("\nFile Content:")

print(content)

except FileNotFoundError:

print("Error: File not found")

except Exception as e:

print("Something went wrong:", e)

Function to write data into file

def write_file(filename, text):

try:

with open(filename, "a") as file:

file.write(text)

```
print("\nData written successfully")
```

```
except Exception as e:
```

```
    print("Error while writing:", e)
```

```
# Function to rename file
```

```
def rename_file(old_name, new_name):
```

```
    try:
```

```
        os.rename(old_name, new_name)
```

```
        print(f"\nFile renamed from {old_name} to {new_name}")
```

```
    except FileNotFoundError:
```

```
        print("Rename Error: File not found")
```

```
except Exception as e:
```

```
    print("Rename Error:", e)
```

```
# Function to move file
```

```
def move_file(source, destination):
```

```
    try:
```

```
        shutil.move(source, destination)
```

```
        print(f"\nFile moved to {destination}")
```

```
    except FileNotFoundError:
```

```
        print("Move Error: File not found")
```

```
except Exception as e:
```

```
    print("Move Error:", e)
```

```
# Function to delete file
```

```

def delete_file(filename):
    try:
        os.remove(filename)
        print(f"\n{filename} deleted successfully")
    except FileNotFoundError:
        print("Delete Error: File not found")
    except Exception as e:
        print("Delete Error:", e)

# ----- MAIN PROGRAM ----- #

print("==== Python File Handling & Automation =====")

# Step 1: Write into file
write_file("sample.txt", "\nWelcome to Python Internship Task")

# Step 2: Read file
read_file("sample.txt")

# Step 3: Rename file
rename_file("sample.txt", "updated_sample.txt")

# Step 4: Create folder if not exists
folder_name = "files"

if not os.path.exists(folder_name):
    os.mkdir(folder_name)
    print("\nFolder created:", folder_name)

# Step 5: Move file into folder
move_file("updated_sample.txt", "files/updated_sample.txt")

```

Step 6: Delete file example

```
delete_file("test.csv")
```

```
print("\nAutomation Task Completed")
```

output:

```
E:\alfido py dev\project>python main.py
===== Python File Handling & Automation =====

Data written successfully

File Content:

Welcome to Python Internship Task

File renamed from sample.txt to updated_sample.txt

Folder created: files

File moved to files/updated_sample.txt
Delete Error: File not found

Automation Task Completed
```

EXPLANATION:

In this task, Python file handling techniques were used to work with text and CSV files. The program reads data from files, writes updated information, and automates common file operations such as renaming, moving, and deleting files automatically using Python modules like `os` and `shutil`.

Exception handling was implemented using `try-except` blocks to prevent the program from crashing when errors occur, such as missing files or invalid file paths. Proper comments were added throughout the code to explain the logic and improve readability.

This task provided practical understanding of:

- File handling in Python
- Automation of repetitive file operations
- Error and exception handling
- Writing clean and understandable code
- Managing files efficiently using Python automation scripts.

Github repository : https://github.com/vineelkaya-create/AlfidoTech_task1/blob/main/main.py